南京邮电大学

实 验 报 告

（2025 / 2026 学年 第 二 学期）

课程名称	C++语言程序设计
实验名称	ARRAYS
实验时间	2026 年 4 月 16 日
指导单位	南京邮电大学计算机学院 数据科学与工程系
指导教师	陈可佳

学生姓名	Alfath Rafiuddin	班级学号	F25040128
学院(系)	Collage of oversees	专 业	Computer Science 1

实 验 报 告

实验名称	ARRAYS			指导教师	陈可佳
实验类型	代码实现	实验学时	2	实验时间	

I TASK

1. Define two arrays for storing characters, copy the characters in the second array to the end of the first array.

```
#include <iostream>
using namespace std;

int main()
{
    char first[] = {'a', 'b', 'c', 'd', 'e'};
    char second[] = {'f', 'g', 'h', 'i', 'j'};

    int size1 = sizeof(first) / sizeof(first[0]);
    int size2 = sizeof(second) / sizeof(second[0]);
    int totalSize = size1 + size2;

    char temp[totalSize];

    for (int i = 0; i < size1; i++)
    {
        temp[i] = first[i];
    }

    for (int i = 0; i < size2; i++)
    {
        temp[size1 + i] = second[i];
    }

    for (int i = 0; i < totalSize; i++) // putting back temporary to first mweheeheh
    {
        first[i] = temp[i];
    }
```

```
	cout << "First Array now is: ";
	for (int i = 0; i < totalSize; i++)
	{
		cout << first[i];
	}

	return 0;
}
// https://youtu.be/6Hk2aE_SRzY?si=bhuhMGYREphfwQ9t use this as reference
```

2. Define a two-dimensional array to store the first ten rows of Yanghui triangle and print it.

```
     1
    1 1
   1 2 1
  1 3 3 1
 1 4 6 4 1
1 5 10 10 5 1
……
```

```
#include <iostream>
using namespace std;

int main()
{
	int pascal[10][10] = {0};
	for (int i = 0; i < 10; i++)
	{
		pascal[i][0] = 1;
		pascal[i][i] = 1;

		for (int j = 1; j < i; j++)
		{
			pascal[i][j] = pascal[i - 1][j - 1] + pascal[i - 1][j];
		}
	}

	// Print Pascal
	for (int i = 0; i < 10; i++)
	{
		// Space, IDK maybe jarak hahaha
		for (int space = 0; space < 10 - i - 1; space++)
		{
			cout << " ";
		}
```

```
        // The number
        for (int j = 0; j <= i; j++)
        {
            cout << pascal[i][j];
            if (j < i)
                cout << " ";
        }
        cout << endl;
    }
    // The output not good enough though, how to center it????
    return 0;
}
```

3. Write a C++ function **int getMaxPrime(int a[], int n)** get the maximum prime from a series of integers if any. Test this function in the **main** function.

```
#include <iostream>
using namespace std;
// Part for check isPrime==true
bool isPrime(int num)
{
    if (num <= 1)
        return false;

    for (int i = 2; i < num; i++)
    {
        if (num % i == 0)
        {
            return false;
        }
    }
    return true;
}
// MAX prime in my array
int getMaxPrime(int a[], int n)
{
    int maxPrime = -1;

    for (int i = 0; i < n; i++)
    {
        if (isPrime(a[i]))
        {
            if (a[i] > maxPrime)
```

```
                {
                        maxPrime = a[i];
                }
            }
        }

        return maxPrime;
}
int main()
{
        int numbers[] = {5, 28, 17, 105, 893, 3, 45, 1};
        int n = 8;

        cout << "Array: ";
        for (int i = 0; i < n; i++)
        {
            cout << numbers[i] << " ";
        }
        cout << endl;

        int result = getMaxPrime(numbers, n);

        if (result != -1)
        {
            cout << "The largest prime number: " << result << endl;
        }
        else
        {
            cout << "There are no prime numbers in the array" << endl;
        }
}
```

指导教师评语						
评分细则	评分项	优秀	良好	中等	合格	不合格
	遵守实验室规章制度					
	学习态度					
	算法思想准备情况					
	程序设计能力					
	解决问题能力					
	课题功能实现情况					
	算法设计合理性					
	算法效能评价					
	回答问题准确度					
	报告书写认真程度					
	内容详实程度					
	文字表达熟练程度					
	其它评价意见					
	本次实验能力达成评价（总成绩）		批阅人		日期	